

192.168.1.0/24 → IP address

31 host

$$2^b = 64 - 2 = 62$$

start: 192.168.1.0/24

Netmask: 255.255.255.0/24 } netmask

192.168.1.0/24

$$\begin{array}{r} 00111101 \rightarrow 62 \\ 00000000 \\ \hline 00111101 \\ 126 \downarrow \\ 111111 \text{ broadcast} \\ 192.168.1.63/26 \end{array}$$

B | 192.168.1.64/28 → base range
192.168.1.79/28 → broadcast

$$2^4 - 2 = 14$$

$$\begin{array}{r} 01000000 \\ 01001110 \\ \hline 01001111 \rightarrow \text{broadcast} \\ 128 \end{array}$$

C | 192.168.1.80/28 → base
192.168.1.95/28 → broadcast
 $2^4 - 2 = 14$

$$\begin{array}{r} 01010000 \\ 01011110 \\ \hline 01011111 \rightarrow \text{broadcast} \end{array}$$

D | 192.168.1.96 →
192.168.1.99 broadcast

$$2^1 - 2 = 2 \quad \begin{array}{r} 01100000 \\ 01100010 \\ \hline 01100011 \rightarrow \text{broadcast} \end{array}$$